

Junwei Quan

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EXPERIENCE

Research Assistant

May 2024 – Apr 2025

York University

Toronto, ON

- Translated research objectives into data-processing and TensorFlow-based image segmentation workflows, including dataset preparation, model evaluation, and analysis of experimental results.
- Built end-to-end Python pipelines to process field measurements, geospatial data, and LiDAR datasets producing structured inputs for machine learning research in remote sensing and vegetation mapping.
- Collaborated with researchers and domain experts to resolve data-quality issues, document methodologies, and communicate findings through technical reports and presentations.

PROJECTS

Agent-Native Collaboration Studio | OpenAI WebMCP Challenge | github.com/Wei-cyber/SynMod

- Owned the product lifecycle from requirements and architecture through implementation, testing, deployment, documentation, and iteration.
- Designed 30+ WebMCP integration tools with typed schemas, structured responses, cancellation support, revision guards, and safe browser confirmations.
- Built a shared command layer and atomic transaction system supporting up to 50 reversible operations, conflict detection, undo/redo, local persistence, and Playwright coverage.

Agentic AI Research & Action Platform | github.com/Wei-cyber/agenticWorkplace

- Built an agentic AI Application in Python that orchestrates LLM reasoning, tool calls, tool execution, and iterative responses through multi-step agent loops.
- Integrated web search and context extraction tools to enable agents to retrieve external information, inspect source content, and incorporate relevant context into responses.
- Developed document upload and processing capabilities as the foundation for retrieval-augmented generation (RAG), enabling the system to ground responses alongside web sources.

Human-Agent Workforce Platform | github.com/Wei-cyber/humanAgentOS

- Developed an AI-driven task-routing prototype that classified work by evaluating key criteria.
- Integrated AWS Bedrock models and tool calling through backend API endpoints, enabling agents to perform multi-step reasoning and invoke application functions during task execution.
- Diagnosed and resolved agent integration issues involving structured tool-result schemas, model invocation, SDK behavior, and tool-step limits during end-to-end testing.

Product Intelligence Platform | github.com/Wei-cyber/productIntelligence

- Built an ETL platform that ingests product pricing and search-ranking data from external APIs via Airflow-scheduled workflows, with built-in data validation, and presents trends through an interactive dashboard supporting pricing analysis, competitive monitoring, and ranking-volatility detection.
- Implemented scheduled workflows using Airflow, API integration, data validation, and analytical visualizations.
- Containerized components with Docker for reproducible deployment and development.

EDUCATION

York University

Toronto, ON

Honours Bachelor of Science in Computer Science

May 2026

- York Student Organization | Marketing Lead

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS

Tools & Platforms: Docker, GitHub, Bash, GitHub Actions

Frameworks: FastAPI, Dash, Airflow

Development Practices: CI/CD, Agile, Scrum